

Interpreting convolutional neural network classifiers applied to laser-induced breakdown optical emission spectra

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ABSTRACT

Laser-induced breakdown spectroscopy (LIBS) is a well-established industrial tool with emerging relevance in high-stakes applications. To achieve its required analytical performance, LIBS is often coupled with advanced pattern-recognition algorithms, including machine learning models. Namely, artificial neural networks (ANNs) have recently become a frequently applied part of LIBS practitioners' toolkit. Nevertheless, ANNs are generally applied in spectroscopy as black-box models, without a real insight into their predictions. Here, we apply various post-hoc interpretation techniques with the aim of understanding the decision-making of convolutional neural networks. Namely, we find synthetic spectra that yield perfect expected classification predictions and denote these spectra class-specific prototype spectra. We investigate the simplest possible convolutional neural network (consisting of a single convolutional and fully connected layers) trained to classify the extended calibration dataset collected for the ChemCam laser-induced breakdown spectroscopy instrument of the Curiosity Mars rover. The trained convolutional neural network predominantly learned meaningful spectroscopic features which correspond to the elements comprising the major oxides found in the calibration targets. In addition, the discrete convolution operation with the learnt filters results in a crude baseline correction.

1. Introduction

Laser-induced breakdown spectroscopy (LIBS) is a prominent atomic emission spectroscopic method [1,2] that enjoys popularity, e.g., in the analysis of industrial [3–5], geological [6–8], and archaeological [9,10] samples owing to the robustness of the LIBS instrumentation and its rapid sampling capabilities. Nonetheless, LIBS spectra are generally characterized by considerable shot-to-shot variability [11,12] and a non-linear signal response [13–15] resulting from the dynamic changes [16–19] and inhomogeneities [20–22] of the optical radiation's source, i.e., the microplasma induced by the laser pulse [16]. Consequently, extensive efforts have been made towards the development of robust data analysis methodologies that can address the non-linear nature of LIBS data [23]. One of the often-used techniques are artificial neural networks (ANNs) [24] which—in the context of LIBS—have been used

for classification [25–30], quantification [24,31,32], dimensionality reduction [33], and baseline correction [34].

Despite their wide-ranging successes, several limitations of ANNs remain, e.g., their tendency of overfitting the training data which leads to poor generalization capabilities when facing newly obtained data [35], and the need for relatively large training datasets. Most notably, ANNs are not directly interpretable [36,37], i.e., they are most often applied as black-box models—especially in fields that are only marginally exposed to machine learning, such as the spectroscopies. This limits the ANNs' trustworthiness, and thus, applicability in high-stakes LIBS applications, e.g., cancer detection which has recently gained traction in the LIBS literature [38–41]. In addition, the understanding of ANN models is highly beneficial for troubleshooting, model optimization, and the design of specialized ANNs. Lastly, LIBS data are often linearly separable owing to their high dimensionality [42]. As such, the

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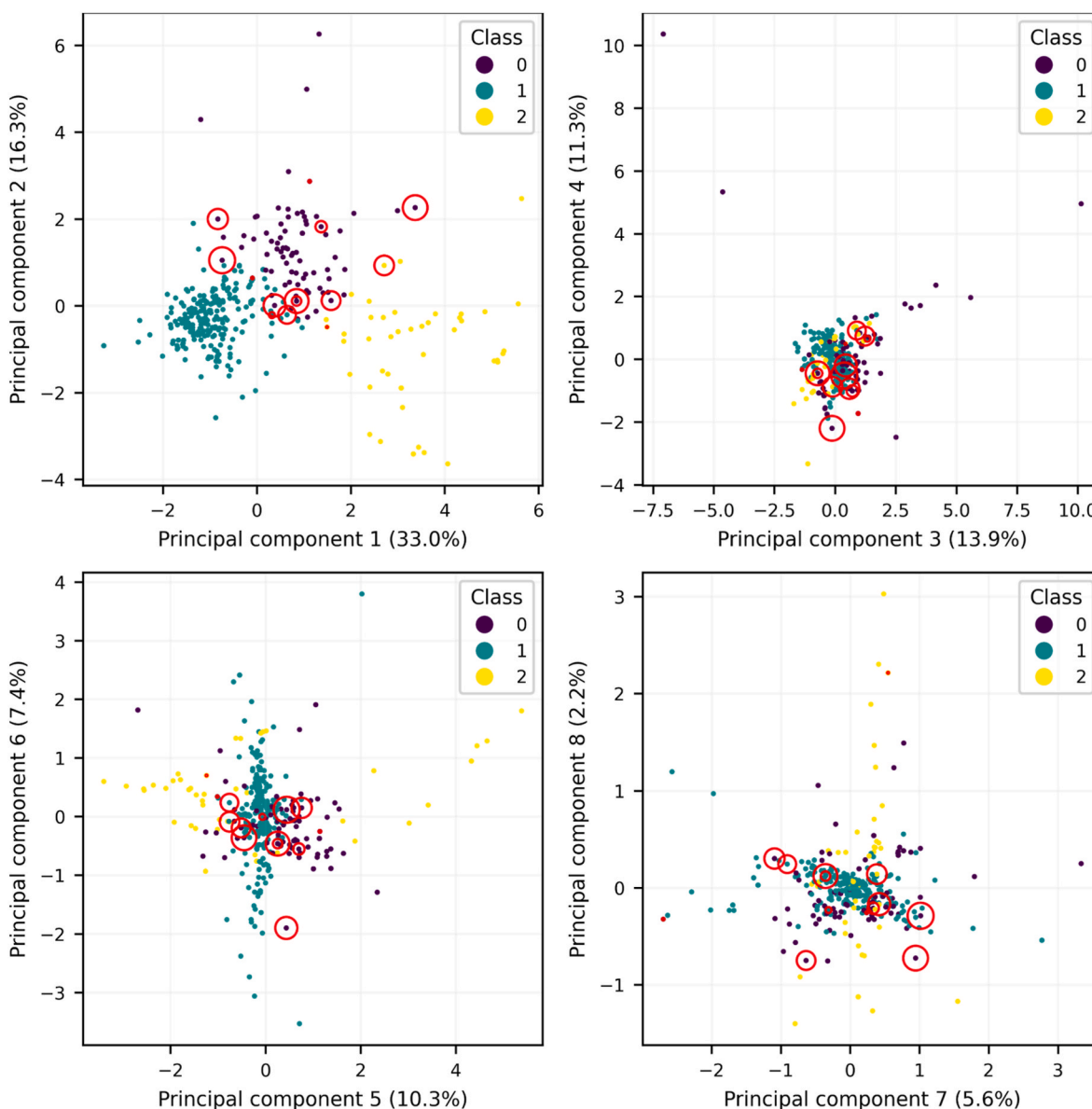


Fig. 1. Principal component analysis scores of the 8 major oxides' concentrations as obtained by reference methods [50]. The colors denote the assigned class labels. The red circles mark the targets whose spectra in $\mathbf{X}_{\text{test}}$ were misclassified. The radius of the red circles marks the number of misclassified spectra of the target (discussed in the Results and Discussion section).

application of complex ANNs is often questionable and linear multivariate models often offer comparable results. Nevertheless, this has been only shown in the context of quantification in low concentration ranges, where LIBS exhibits linear signal response [43].

This prompts the development of a methodology that can be used to interpret ANNs applied to spectroscopic data, such as LIBS spectra. Nevertheless, there have only been limited efforts made towards interpreting ANNs applied for LIBS, e.g., by visualizing the t-distributed stochastic neighbour embedding [44] of the dataset as it was transformed by a trained ANN's layers [45]. More recently, this approach was augmented by the application of the gradient-weighted class activation mapping [46] for the interpretation of ANN models used for classifying lab-grown tumour cells [47]. More commonly, the classification performed via ANN is interpreted using other, linear models [48].

In this work, we demonstrate several tools applied to interpret the decision-making of convolutional neural networks (CNNs) trained to classify the extended ChemCam [49] calibration dataset [50]. While the use of fully connected ANNs is currently prevalent in the LIBS literature,

CNNs offer attractive feature extraction capabilities [51], as has been recently demonstrated in various LIBS applications, e.g., in the context of soil sample classification [52,53], the quantitative analysis of the extended ChemCam calibration dataset [54,55], and aluminium scrap classification [56]. Nevertheless, the overall dominance of CNNs over fully connected neural networks has been recently questioned in the small data regime [57]. In addition, CNNs generally comprise several hidden layers, and thus, belong to the family of deep learning models. Deep learning models are significantly overparameterized, i.e., they possess (often orders of magnitude) more trainable parameters than the number of observations used to train them. Consequently, the CNN limitations might be more prone to overfitting than their fully connected counterparts. Nevertheless, CNNs have been recently shown to exhibit implicit self-regularization, which improves their ability to generalize better [58].

Here, a simple CNN architecture is trained to classify spectra of 380 homogenized geological material targets sampled by the ChemCam testbed mast unit [50]. The trained model is interpreted post-hoc, i.e.,

after its training is completed. Input spectra are found which maximize mean of the individual feature maps in the network and the activation of the neurons in the last (softmax) layer carrying out the classification. The reproducibility of obtaining these spectra is also investigated. Lastly, the shape of the convolutional filters is explored and their impact on the input spectra visualized.

2. Materials and methods

2.1. Data

2.1.1. Acquisition

The explored CNN was trained to classify the part of the extended ChemCam LIBS calibration dataset [49,50,59,60]. The dataset contained spectra of 380 homogenized and pelletized geological samples, obtained from Refs. [61,62]. The spectra were measured under a 7 ± 0.2 Torr CO₂ atmosphere using a Nd:KRG diode-pumped laser with a wavelength of 1067 nm, 14 mJ pulse energy (1.5 GW/cm² flux), and 5 ns pulse length at a distance of 1.5 m. The plasma emission was resolved and recorded in the 240–905 nm wavelength range using three Czerny–Turner spectrographs each equipped with a back-illuminated CCD detector. Each target was sampled at five distinct spots. From each spot, 50 spectra were collected. Each of these spectra represents a distinct data point in the current work. The mean spectrum of the dataset and its standard deviation are shown in Fig. S1 (Supplementary materials).

2.2. Label creation and train-test split

The dataset was constructed with the aim of calibrating regression models. Hence, no classification labels are available for every target, such as the geological type of the source material. Consequently, the labels were created from the chemical composition of the targets *via* the following process.

- 1) The Z-scores of the 8 major oxides were calculated (by mean centering their concentrations in wt.% and scaling them to unit variance). Z-score refers to a value's distance from the sample's mean in terms of standard deviations;
- 2) Principal component analysis (PCA) [63] of the compositions' Z-scores was performed (Fig. 1) using the Python programming language's scikit-learn library (*via* singular value decomposition);

3) The PCA scores were clustered using the k-means algorithm with the Euclidian distance metric. Three clusters were considered, according to Fig. 1. The validity of this choice was also confirmed using the silhouette score for clusters in the 2–15 range. The chosen cluster count of 3 resulted in silhouette score of 0.3444 ± 0.0018 (with 1 corresponding to a “perfect” clustering and –1 corresponding to completely overlapping clusters). This relatively low silhouette score is mainly the result of the unclear boundary between the clusters. Nevertheless, the low uncertainty of the achieved silhouette score shows that the clusters are identified in a repeatable manner.

The resulting clustering is shown in Fig. 1. It is important to point out that the number of identified clusters differs (3) from the number of oxides (8). While the targets are composed of 8 major oxides, SiO₂ and Al₂O₃ make up most of the targets, namely 54% and 14% on average, respectively. In comparison, the remaining 6 oxides represent, on average, less than 5%. Thus, the clusters identified *via* the clustering of the chemical composition of the targets are more complex than simply considering the most prominent oxide in the given target. Subsequently, the cluster labels were used as class labels for the consequent classification. Here we emphasize that the clustering was performed on the target compositions, while the classification was done on LIBS spectra. Hence, the $N_S = 94950$ spectra of the 380 targets were distributed among $N_{class} = 3$. Subsequently, the dataset $\mathbf{X} \in \mathbb{R}^{N_S \times N_{wvl}}$ (where $N_{wvl} =$

6144 is the number of independent variables, i.e., the number of resolved wavelengths and the size of the model's input) was randomly split into $\mathbf{X}_{train} \in \mathbb{R}^{68500 \times N_{wvl}}$, $\mathbf{X}_{validation} \in \mathbb{R}^{13150 \times N_{wvl}}$, and $\mathbf{X}_{test} \in \mathbb{R}^{13300 \times N_{wvl}}$ for the training, validation, and testing of the CNNs, respectively. The spectra in the three datasets originate from three mutually exclusive sets of targets. The number of spectra corresponding to the individual classes in the three datasets are 15000–44500–9000 in $\mathbf{X}_{train}$ (corresponding to approximately 71%–72%–75% of the available data for the three individual classes), 2950–8750–1450 in $\mathbf{X}_{validation}$ (14%–14%–12%), and 3050–8750–1500 in $\mathbf{X}_{testing}$ (15%–14%–13%). As such, the considered classification task is imbalanced, with Class 1 comprising the bulk of the dataset, namely 3- and 5-times more targets assigned to Class 1 than to Class 0 and Class 2, respectively. Consequently, apart from the frequently used classification accuracy, precision and recall are also reported.

2.3. Spectral pre-processing

Each spectrum was individually divided by the maximum intensity of the training dataset. This was done to maintain the relative intensity difference among spectra of distinct samples, while scaling all intensity values approximately to the 0–1 range. In contrast, by scaling each spectrum individually, information about the relative intensity differences would be lost. No further pre-processing steps were employed.

2.4. Convolutional neural networks

Artificial neural networks are mathematical models $f_\theta(\cdot)$ which take an input $\mathbf{x}$ and yield a prediction $\hat{\mathbf{y}}$, i.e., $f_\theta(\mathbf{x}) = \hat{\mathbf{y}}$. ANNs consist of nodes (artificial neurons) arranged into layers [64]. In their most basic form, the neurons of an ANN are connected to every neuron in both adjacent layers (except for the input and output layers, which have a single adjacent layer); this architecture is commonly referred to as a fully connected ANN (or multilayer perceptron) [65]. A fully connected neuron takes the weighted sum of the output of every neuron in the preceding layer and maps it onto a scalar *via* the activation function.

In contrast, convolutional neural networks (CNNs) contain at least one convolutional layer [66–68]. The neurons in a convolutional layer perform the discrete convolution operation on the output of the preceding layer, which—after the application of the activation function—can be written as:

$$a_{i,k}^{(l)} = \sigma \left(\sum_{j=0}^J w_{j,k}^{(l)} \cdot a_{i+j}^{(l-1)} + b_k^{(l)} \right) \quad (1)$$

Here, $a_{i,k}^{(l)}$ is the i^{th} vector element of the k^{th} convolutional neuron's output in the l^{th} layer, $w_{j,k}^{(l)}$ are the corresponding weights, $b_k^{(l)}$ is a corresponding bias term; and $\sigma(\cdot)$ is the activation function [69,70]: a non-linear mapping $\mathbb{R}^1 \rightarrow \mathbb{R}^1$, i.e., a function which transforms the sum's value in a non-linear manner. Here, we used the rectified linear unit (ReLU) activation function defined as $\sigma_{ReLU}(u) = \max(0, u)$. The weights $w_{j,k}^{(l)}$ of a convolutional neuron are often referred to as the shape of the convolutional filter.

We explored several convolutional neural network architectures, ranging from a single convolutional layer to up to 4 convolutional layers combined with max pooling layers (Fig. S2–S4, in supplementary materials). However, the larger models (the mid-sized and the largest model) did not outperform the smallest convolutional neural network. As such, we performed the subsequent steps only considering the smallest explored model.

The trained model consisted of a single convolutional layer with $N_f = 16$ neurons (distinct convolutional filters). Each convolutional filter was $J = 8$ units wide. The output of each convolutional neuron was a vector of length $(N_{wvl} - J + 1)$ since no padding and a stride of 1 were used. Considering a training dataset of $\mathbb{R}^{N_S \times N_{wvl}}$, the output of the con-

volitional layer was a three-dimensional array with dimensions of $\mathbb{R}^{N_c \times (N_{wvl} - J + 1) \times N_f}$. The convolutional layer was followed by a flattening layer which aggregates the vectors from the N_f channels into a single vector of length $N_f \cdot (N_{wvl} - J + 1)$, i.e., unfolds the transformed data cube. Lastly, the flattening layer was followed by a softmax layer with $N_{class} = 3$ neurons, i.e., a single softmax neuron corresponding to each class in the dataset. Each softmax neuron applies the exponential function to the dot product of its input vector of length $N_f \cdot (N_{wvl} - J + 1)$ and the neuron's input weights and outputs single scalar. Consequently, the outputs of the N_{class} softmax neurons are normalized to unit sum. Thus, the output of the softmax neurons can be interpreted as the probability (or prediction confidence) of the predicted input's class label.

Both the weights $w_{j,k}^{(l)}$ and bias terms $b_k^{(l)}$ are learnt *via* gradient descent with backpropagation [71] during training in a supervised manner: each training data point x is assigned an expected class label y (expressed as a vector *via* one-hot encoding). Consequently, the model is used to predict the class label $\hat{y}$, which is compared to the expected class label y . Then, the difference between y and $\hat{y}$ is expressed using a loss function $\mathcal{L}(\theta, y, \hat{y})$, where θ is a vector of the model's weights and bias terms. Lastly, each of the model's weights θ_i is iteratively adjusted by a value proportional to $-\frac{\partial}{\partial \theta_i} \mathcal{L}(\theta, y, \hat{y})$, i.e., through gradient descent. We used the categorical cross entropy loss function [72], which is a common choice for classification tasks. Our CNN was trained using the Adam optimizer [73] with a minibatch size of 16 for 5 epochs and an initial learning rate of $\eta = 3 \cdot 10^{-4}$.

2.5. Interpretability techniques

Most machine learning (ML) models are not directly interpretable, i.e., they do not yield a closed-form description of the function they approximate. Consequently, various post-hoc interpretability techniques were developed to validate trained neural networks [36,74,75]. These range from simply inspecting the model's weights to finding prototype inputs (or prototype spectra; we use these two terms interchangeably) which yield a desired prediction. In the present work, the following techniques are employed.

- The convolutional filters' shapes are extracted from the trained model. Inspecting the learnt convolutional filter shapes reveals what the CNN is looking for.
- The data points' transformation by the discrete convolution using the learnt filters is explored: the discrete convolution of the extracted filters and measured spectra is performed. Understanding what happens to the spectra while being processed by the CNN can show whether the network enhances undesired—or unexpected—spectral features.
- Class-specific optimal spectra x_c are found. These prototype spectra yield perfect predictions for a single class, i.e., a perfect one-hot encoding vector. The class-specific optimal spectra are found by the gradient descent approach described above. However, instead of adjusting the weights, a randomly initialized input spectrum is iteratively adjusted to minimize the prediction error:

$$x_c = \operatorname{argmin}_x \mathcal{L}(x, y_c, \hat{y}) \quad (2)$$

while keeping the model's weights intact. The spectra x_c represent the model's expectation for a given class c [76,77]. This approach can be slightly modified to find spectra that maximize a single feature map, by taking the mean or sum of the convolutional neuron's feature map as the loss function. Lastly, note that various regularization terms can be added to $\mathcal{L}(\cdot)$ in (2), e.g., to obtain a sparse solution. This is desirable since commonly LIBS spectra are parsimonious [78]. Here, we considered the L_1 and L_2 norms ($\sum |x_{c,i}|$ and $\sum x_{c,i}^2$, respectively, where $x_{c,i}$ are the vector components of x_c). In addition, the total absolute difference of the adjacent values of x_c was also considered [76], which is henceforth

referred to as the total variance $L_{TV} = \sum |x_{c,i} - x_{c,i+1}|$ [79,80]. The impact of the regularization terms can be adjusted using the corresponding scalar coefficients λ_{L1} , λ_{L2} , and λ_{LTV} . The aim of applying the L_1 and L_2 regularizations is to bound the intensities yielded by the optimization process (similarly to the use of these regularization terms in multivariate regression problems, where their application limits the magnitude of the individual regression parameters). Consequently, by using L_1 and L_2 regularizations, we attempt to force the optimizer to rely on producing prototype spectra which resemble measured spectra, that is, are composed of well-resolved emission lines. The total variance regularization is included to further promote the production of realistic spectra by suppressing the sudden and disproportional change in adjacent intensity values.

To optimize the values of λ_{L1} , λ_{L2} , and λ_{LTV} , two criteria were considered: 1) the yielded prototype spectra should be classified with perfect confidence (i.e., the CNN's prediction on the prototype spectra should be a perfect one-hot encoding vector) with a chosen tolerance of < 0.01 ; 2) the prototype spectra should contain the lowest possible number of non-zero intensity values to facilitate their interpretation. These criteria were fulfilled by manually adjusting the λ parameters without an extensive and exhaustive optimization. An automatic approach will be developed in a future work.

The question of whether the found prototype spectra comprise the emission lines responsible for the dataset's classification was explored. Firstly, PCA was performed on the full dataset and the scores of the first 8 principal component scores were maintained. Secondly, the k -nearest neighbour classification (kNN) [81] of the scores was performed using the Euclidean metric and the $k = 250$ nearest neighbours (the number of spectra obtained from each target). As distance metrics often exhibit unexpected behaviour in high dimensions (such as LIBS spectra consisting of thousands of resolved wavelengths) [82,83], we performed PCA prior to the kNN classification. Subsequently, the kNN classification was performed on the PCA scores. This classification performance formed a baseline and is denoted as kNN_{full}. Subsequently, the dataset was truncated: only wavelengths with a corresponding intensity value above 0.05 in at least one of the prototype spectra were kept. The truncated dataset was then classified according to the steps above, i.e., PCA followed by kNN. The corresponding results are denoted as kNN_{truncated}. Here, kNN is used instead of retraining the CNN because truncating the wavelength range is expected to disturb the local patterns exhibited by the data (e.g., emission lines). Here we emphasize that classification using kNN models is only carried out to verify that the wavelength ranges present in the prototype spectra are sufficient for classifying the dataset. Hence, the kNN performance is not to be compared with that of the CNN models. Lastly, we opted for kNN over other multivariate classification techniques because kNN considers only the local neighbourhood of the observations. That is, kNN does not transform the dataset prior to predicting the class memberships. As such, kNN allows us to empirically validate whether truncating the spectra changes the local neighbourhood of the observations or not. Note that the linear transformation performed by PCA does not affect the local neighbourhood either.

Lastly, for the exploration of the features found as important by the CNN, the common partial least squares discriminant analysis (PLS-DA) was used as reference. PLS-DA performs a linear dimensionality reduction, similar to that of PCA. However, in contrast to PCA, the PLS-DA dimensionality reduction is guided by the ground truth information, i.e., PLS-DA is trained in a supervised manner, while PCA is a completely unsupervised technique which must be combined with a classification algorithm for material identification. Consequently, the loading (or weight) vectors yielded by PLS-DA (and PLS regression) are often used to identify important features in the classified dataset [53,84]. The calculations have been done using the Python programming language (v3.9.16).

The neural networks and their interpretations have been

Table 1

Summary of the achieved classification performances on the three datasets. CNN refers to the mean of the 3 trained CNN models with the uncertainty being the standard deviation. kNN_{full} and $kNN_{truncated}$ denote to the classification of the 8 first principal components' scores obtained from the full or truncated datasets, respectively. Truncation refers to the selection of the wavelength ranges with corresponding intensity values above the chosen threshold of 0.05 in at least one of the prototype spectra. The thresholding was performed individually for the three trained CNNs. The uncertainty is the standard deviation of the resulting 3 classifications.

Dataset	Classifier	Balanced Accuracy [†]	Precision	Recall
X_{train}	CNN	0.942 ± 0.013	0.940 ± 0.010	0.930 ± 0.010
	kNN_{full}	0.963	0.952	0.965
	$kNN_{truncated}$	0.969 ± 0.007	0.959 ± 0.009	0.971 ± 0.003
$X_{validation}$	CNN	0.847 ± 0.023	0.890 ± 0.010	0.880 ± 0.020
	kNN_{full}	0.779	0.662	0.822
	$kNN_{truncated}$	0.83 ± 0.040	0.732 ± 0.063	0.865 ± 0.015
X_{test}	CNN	0.869 ± 0.015	0.900 ± 0.020	0.890 ± 0.020
	kNN_{full}	0.851	0.807	0.817
	$kNN_{truncated}$	0.825 ± 0.017	0.753 ± 0.027	0.830 ± 0.015

implemented using the Keras 2.11.0 [85] library running on the Tensorflow 2.11.0 [86] backend. The PCA and clustering have been performed using the Scikit-learn 1.2.1 library [87]. The work has been done in the Google Colaboratory cloud environment (free version). The CNNs have been trained on central processing units (CPU) without accelerators (such as graphical processing units). Nevertheless, the training took up to several minutes.

3. Results and Discussion

The considered architecture was trained and evaluated 3 times on the same datasets with three distinctly initialized sets of weights. Henceforth, these three models are distinguished with indices 0–2. Nevertheless, we refer to the same model architecture. During training, the model's performance on $X_{validation}$ decreased immediately after the first epoch, i.e., subsequent epochs lead to the model's overtraining. Hence, in the following, the models' state after the first epoch is considered. The performance of the trained CNNs is summarized in Table 1 (the corresponding confusion matrices are shown in Fig. S5–S7, Supplementary materials). Despite the early stopping, the models' performance on X_{train} was still higher than their test performance. On the contrary, the models' validation performance was indicative of their test performance. Here, we note that increasing the model complexity (adding additional layers and increasing the layer sizes) did not improve the validation performance. Hence, these efforts are not presented here and the impact of modifying the CNN architecture will be explored in detail in our future work. For reference, the linear PLS-DA classifier with 18 latent variables (optimized via standard cross-validation) yielded a test balanced classification accuracy of 0.852.

The test targets whose spectra were misclassified are highlighted in Fig. 1. Notably, the targets with the highest number of misclassified spectra lay on the perceived class boundaries in the composition principal component scores space. Thus, the correct classification of these spectra would require the model to extrapolate, which is, in fact, the most challenging situation for identification models. This, without the insertion of implicit domain knowledge into the network architecture or

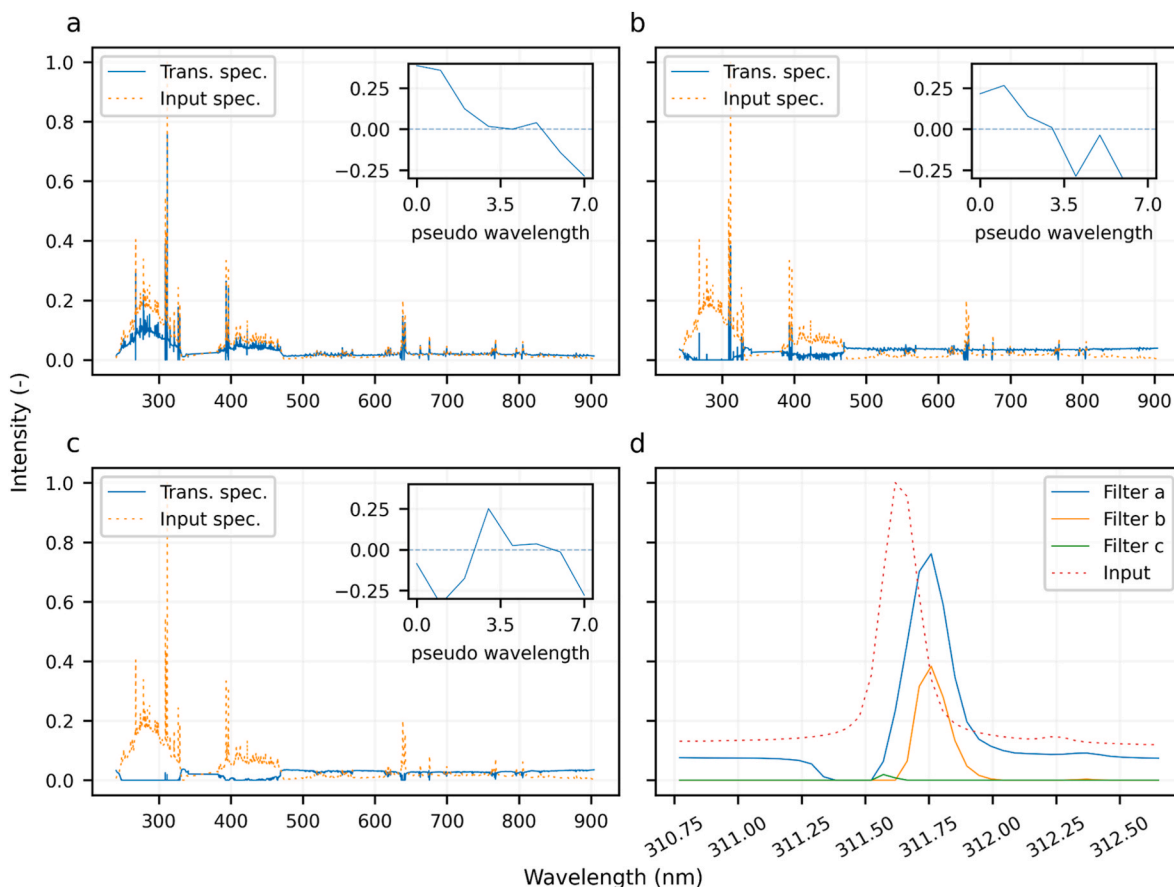


Fig. 2. Examples of the convolutional transformation of a measured spectrum by three distinct filters learnt by the CNN in its first layer (a–c) and the transformation of a single emission line by these three convolutional filters. The insets show the corresponding filter's shape that yielded the spectrum's shown transformation. d) shows in detail the spectrum's strongest emission line's transformation by the selected 3 convolutional filters. Pseudo wavelength refers to a single ordinal data index. The insets' y axis shows unitless weight values. Trans. Spec – Transformed spectrum; Input spec. – Input spectrum, i.e., spectrum prior to the transformation.

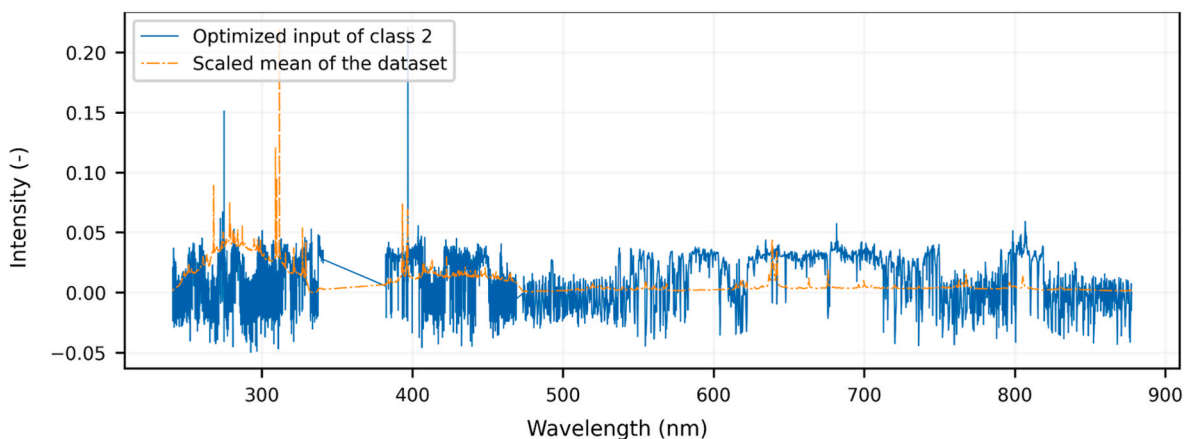


Fig. 3. Prototype spectrum yielding a perfect prediction for class 2 obtained by unregularized gradient descent.

training process, is a well-known limitation of data-based learning approaches [88]. Considering the model's adequate performance on spectra of targets which did not require extrapolation, it is expected that the trained CNNs learnt to recognize spectroscopic features.

[†]Balanced accuracy is defined as $\frac{1}{2} \left(\frac{TP}{P} + \frac{TN}{N} \right)$ [89], where TP, TN, P, and N stand for true positives, true negatives, positives, and negatives, respectively. The balanced accuracy is calculated for each class separately from a 2×2 confusion matrix and averaged to obtain the balanced accuracy of a single model. The values reported in the table refer to the mean balanced accuracy of the 3 models (where applicable). The uncertainty refers to the corresponding standard deviation.

In fact, as shown in Fig. 2a–c (insets), the learnt convolutional filters resemble portions of emission lines, albeit noisy ones. The negative values of the convolutional filters suggest that they act as edge detectors [90]: they magnify the ascending or descending sides of the emission line. Consequently, the emission lines in the output vector of a convolutional neuron are wavelength-shifted, as shown in Fig. 2d. The direction of the shift depends on the slope of the filter's shape.

An additional consequence of the discrete convolution is the apparent baseline correction (Fig. 2b). Spectral regions that negatively or weakly correlate with the filter's shape yield negative values (owing to a negative bias term) and are consequently set to zero by the chosen ReLU activation function that has strictly non-negative outputs. While baseline correction has been reported using both convolutional [55] and fully connected neural networks [34], in both cases the network has been explicitly trained to do so. That is, the network was trained to predict the baseline-corrected spectrum from the raw measured

spectrum. In addition, the convolutional neural network used to perform baseline correction was significantly larger than our model; it consisted of 20 convolutional layers with a kernel size of 3 [48]. On the contrary, our results show that a single convolutional layer (although with a larger kernel size) can suppress the spectral baseline. Nevertheless, not every filter act in such a beneficial manner: the filter shown in Fig. 2a appears to have negligible overall effect on the spectrum apart from introducing a few spectral artefacts. Meanwhile, the filter in Fig. 2c appears to flatten the spectrum apart from a small baseline above 480 nm.

Overall, the filter shapes and the corresponding transformations of the input data by the convolutional layer provide limited insight. Instead, the exploration of the features considered to be important by the trained model is more informative. Thus, class-specific prototype spectra were found via input optimization by gradient descent (Fig. 3). However, the prototype spectra shown in Fig. 3 are highly noisy, which is a known limitation of the input optimization approach [37]. These optimal inputs are commonly considered adversarial examples, i.e., data points that yield a desired prediction while not bearing any resemblance to the class's other data points [91]. Consequently, while the found prototype spectra are predicted with perfect confidence and they exhibit spectroscopic features, identifying the important features based on these prototype spectra is still challenging.

To understand the source of the noise, a gradient ascent approach was used to find input spectra that maximize the individual convolutional neurons' feature map's mean (Fig. 4). The resulting prototype spectra qualitatively corresponds to the noise observed in Fig. 3, i.e., the noise is the superposition of input spectra which maximize the

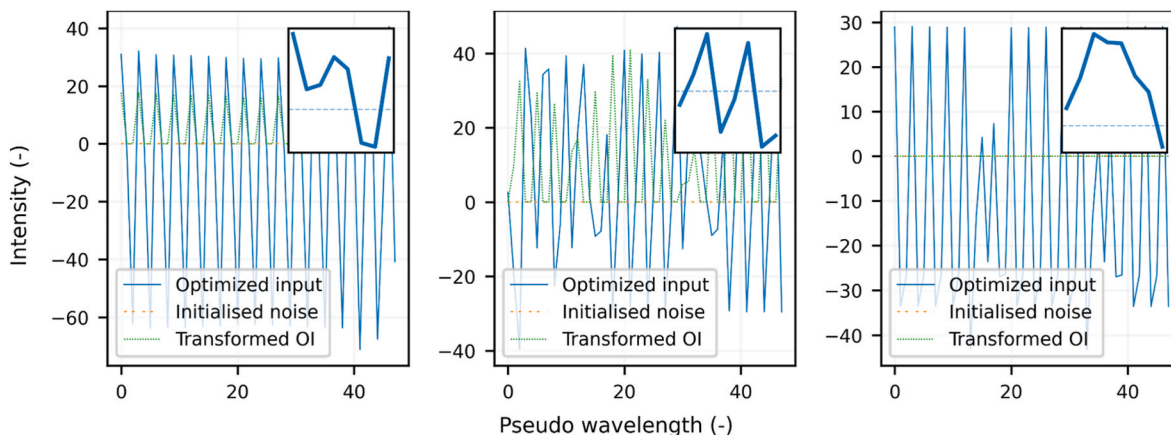


Fig. 4. Portions of the spectra maximizing the feature map's mean for three filters (shown in the insets) of the trained CNN's first (and only) convolutional layer. The dotted lines show the transformation of the optimal inputs by the corresponding filter. The dashed line represents the initial noise $\sim \mathcal{N}(0, 10^{-3})$ from which the spectra were obtained via gradient ascent. OI – optimal input.

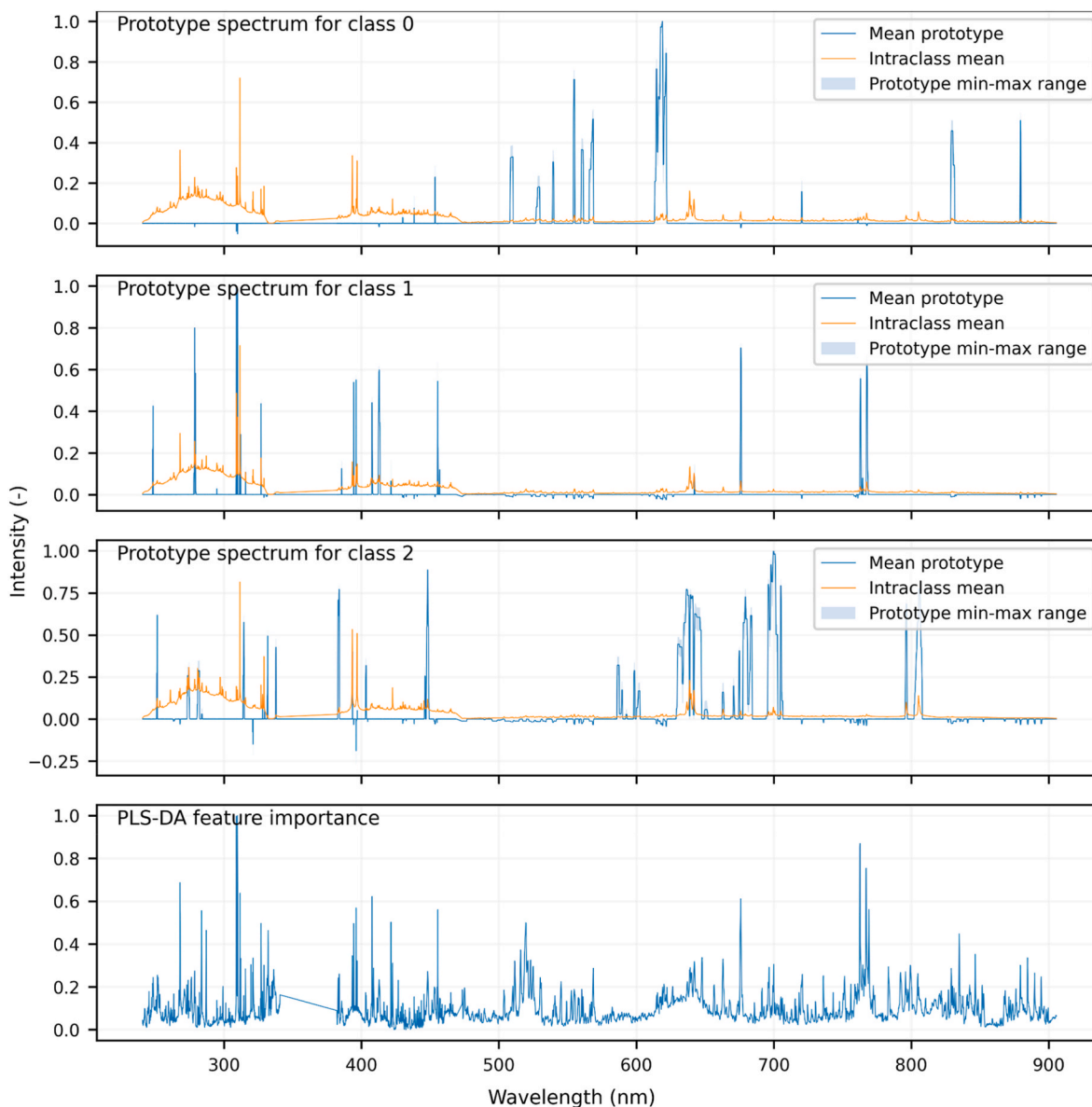


Fig. 5. Examples of prototype spectra (first 3 panels) yielding optimal predictions obtained by regularized gradient descent. By intraclass mean we refer to the mean of the spectra belonging to the given class. The prototype and intraclass mean spectra were scaled to unit maxima. The shaded area shows the min-max range of the prototype spectra obtained from 4 distinct noise vectors (spanning 4 orders of magnitude). The bottom panel shows the feature importance as determined by the PLS-DA classifier.

convolutional neurons with distinct kernel shapes. Consequently, to facilitate the interpretation of the obtained class-specific input spectra, the contribution maximizing the activation of the convolutional neurons should be removed—e.g., by deconvolution—from the optimal input spectra. However, the filters' shapes and the filter maximizing inputs are not smooth. Thus, the deconvolution attempts lead to numerical instabilities.

Therefore, to suppress the noise resulting from the convolutional neurons, the optimization used to find the class-specific representative spectra was repeated including regularization (with $\lambda_{L1} = 0.02$, $\lambda_{L2} = 0.005$, and $\lambda_{LTV} = 0.001$). The chosen parameters λ_{L1} , λ_{L2} , λ_{LTV} were optimized qualitatively, taking into account the approximately perfect prediction confidence on the obtained prototype spectra, the prominence of the noise originating from the convolutional neurons, and the sparsity of the yielded prototype spectra. In comparison with the unregularized class-specific prototype spectrum (Fig. 3), the regularized prototypes (Fig. 5) exhibit no noise and very few emission lines, i.e., parsimonious class-specific prototype spectra were found, which is in

line with the common nature of LIBS spectra [78].

Meanwhile, the regularization had no detectable negative impact on the prediction confidence (values of 0.9944 ± 0.0016 were obtained, i. e., no difference was observed up to the third decimal point compared to the unregularized prototype spectra). For reference, the prediction confidence achieved for the correctly classified training data was 0.92 ± 0.12 . This suggests that the neuron activations induced by most of the emission lines (or noise) present in the unregularized prototype spectra destructively interfere, i.e., the noise's impact is cancelled out at the softmax layer. Instead, the optimization's regularization enforces the elimination of these interfering features while maintaining the spectral features that contribute to the class's identification. We subsequently explored the stability of the prototype spectra by repeating the optimization process starting with a distinct random vector spanning 4 orders of magnitude. The resulting prototype spectra were qualitatively the same (Fig. 5), showing only negligible quantitative variations.

As a reference, we explored the feature importances as determined by a PLS-DA classifier (Fig. 5, bottom panel), whose number of latent

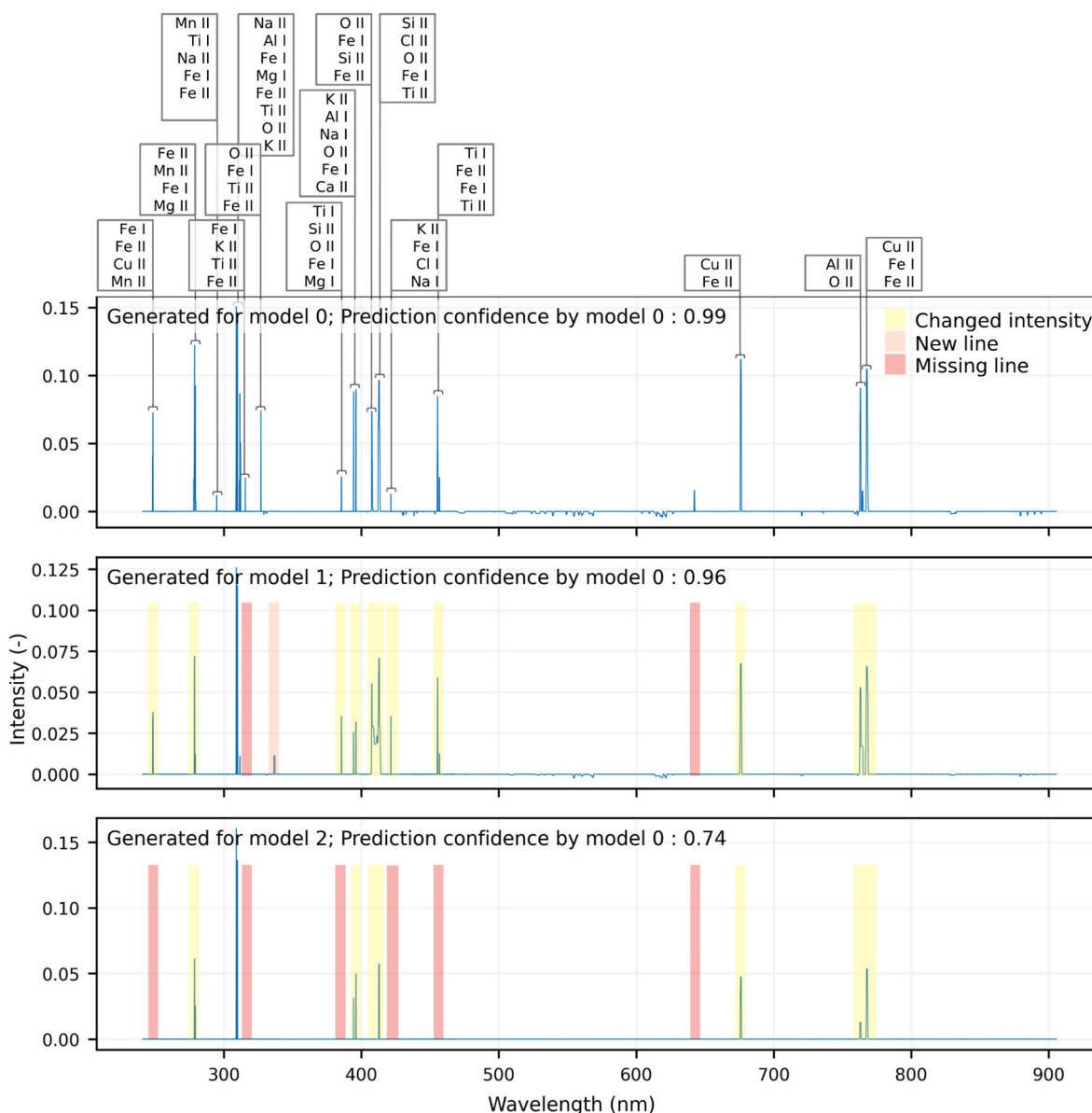


Fig. 6. Prototype spectra of Class 1 obtained for the 3 differently initialized but otherwise identical models. The top panel lists the species whose emission lines correspond to the features present in the prototype spectra. A detailed list is given in [Table S1 \(Supplementary materials\)](#).

variables was optimized using a standard cross-validation process. A major difference is that the PLS-DA weights are not class-specific. Thus, the PLS-DA weights offer less information about the interclass differences. In addition, the PLS-DA weights cover the whole wavelength range of the spectra (i.e., they aren't sparse), making line identification more tedious. This could be addressed by applying a sparse PLS solution. This, however, requires the optimization of further hyperparameters [78] and was, therefore, not explored here. On the contrary, many of the strong emission lines present in the PLS-DA weight spectrum are also present in at least one of the prototype spectra. Namely, the regions around 315 and 760 nm appear strongly in the prototype spectrum of Class 1 (which is the most populous class). The most intense spectral regions in the prototype spectra of Classes 0 and 2 (around 620 and 700 nm), however, appear with lower intensities in the PLS-DA weight vector.

Nevertheless, the prototype spectra were not completely interchangeable among the three trained CNNs, despite the models possessing the same architecture (but trained with distinct initializations). Namely the classification confidence for a prototype spectrum obtained

for model 0 dropped to 0.76 (but still predicted the correct class label) when the prediction was carried out by model 2. Hence, the same architecture trained on the same dataset can learn to weight the same features differently. This is demonstrated in [Fig. 6](#), which shows the prototype spectrum obtained for Class 1 for the three models.

By taking the prototype spectrum obtained for model 0 as reference, the prototype spectra obtained for models 1 and 2 differ in two primary ways: 1) The relative intensity of most lines changes among the prototype spectra (highlighted by yellow in [Fig. 6](#)); and 2) multiple lines disappear (highlighted by red in [Fig. 6](#)). In addition, in the case of the prototype spectrum obtained for model 1, a single additional emission line can be observed (highlighted by green in [Fig. 6](#)). Among the emission lines present in the prototype spectrum of model 0, only a cluster of emission lines between 309 nm and 310 nm remain unchanged. This spectral region corresponds to atomic or ionic emission lines of most elements included in the tables, as listed in the top portion of [Fig. 6](#) (a detailed list is included in [Table S1](#), Supplementary materials, identified using the NIST emission line database [92]).

The class-specific prototype spectra reveal several aspects of the

CNN's decision-making. For example, the emission lines found in the prototype spectrum of Class 1 correspond to Fe I, Fe II, Ti I, Ti II, K I, K II, Mn I, Mn II, Ni I, Cu I, Cu II, and Si I (Fig. 6), which are all emission lines of elements forming the major oxides comprising the targets or minor elements present in some of the targets. Consequently, it can be concluded that the classification is done based on spectroscopic features, rather than, e.g., unique noise patterns.

Finally, the possible interpretation of the class-specific prototype spectra was explored. It would be reasonable to assume that the emission lines present in the prototype spectra are sufficient for the classification of the dataset. Consequently, the datasets were truncated according to the intensity of the prototype spectra. Then, PCA was applied for dimensionality reduction and the PCA scores were classified using kNN. The comparison of the kNN performance on the truncated and full datasets is shown in Table 1: no noticeable difference can be seen on X_{train} , which is expected. More importantly, the observed changes in classification performance on $X_{\text{validation}}$ and X_{test} are within the obtained uncertainties. We further verified this claim by applying the sum of ranked differences algorithm [93] using the program available online [94]. We used the models' balanced accuracy, recall, and precision on the three datasets for creating the ranking. Indeed, the performance of the kNN models was indistinguishable (while the CNN outperformed both). Thus, the model-specific important features correspond to the data-specific ones [42]. However, no clear correlation was found between the emission line intensities (line maxima nor line areas) in the prototype spectra and their feature importance (impact on the classification accuracy).

4. Conclusion

A simple convolutional neural network, consisting of a single convolutional and a single fully connected layer was trained to classify the dataset constructed to calibrate the Curiosity Mars rover's ChemCam LIBS instrument. The ground truth values for the classification were created by clustering the targets' compositions. The yielded classification task exceeds commonly reported classification tasks in difficulty, where the training and testing data are essentially sampled from the same multidimensional Gaussian distribution. Nevertheless, the model—despite its simplicity—performed relatively well and was interpreted post-hoc by finding synthetic spectra that are representative of the individual classes. The simple gradient descent approach was found to yield adversarial examples which did not contribute to the interpretability of the CNN. However, an appropriately regularized gradient descent optimization yields class-specific prototype spectra that contain well-defined emission lines. These emission lines are sufficient for classifying the dataset. Thus, trained convolutional neural networks preferentially rely on spectroscopic features to carry out their decisions.

Credit authors statement

EK: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Writing - original draft. **JV:** Conceptualization, Methodology, Writing - review & editing. **TB:** Supervision, Writing - review & editing. **PH:** Supervision, Writing - review & editing. **PP:** Writing - review & editing, Project administration. **JK:** Writing - review & editing, Funding acquisition.

Declaration of competing interest

There are no conflicts to declare.

Data availability

We used publicly available data.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.talanta.2023.124946>.

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